

Multi-Channel Ad Spend Allocation

Under Uncertainty

Thompson Sampling and time-series ML for real-time performance-marketing budget optimization

Abdullah Çopur

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■ The Economic Problem

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Performance marketing is a recurring capital-allocation problem under uncertainty: each day, deploy a fixed ad budget across placements $\times$ creatives $\times$ audiences $\times$ hours.

01

Information asymmetry

Meta pixel attribution overcredits parallel campaigns. Four parallel campaigns for one brand each showed ROAS 80+ on double-counted revenue.

02

Creative fatigue as depreciating capital

Marginal product of a creative declines with cumulative impressions. Without a fatigue forecast, depreciated capital is held longer than optimal.

03

Channel substitution under unknown returns

Audience Network absorbed $\approx 1.8\%$ of total platform spend at 0 attributed revenue. The naïve operator never tests the withdrawal counterfactual.

04

Decision latency vs. operator agency

Faster automation raises the cost of false positives. A binding constraint: no Meta API mutation without per-action operator approval.

Maps cleanly onto principal-agent, multi-armed bandit, and depreciation-of-intangible-capital literatures.

■ Research Question & Data

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RESEARCH QUESTION

Can a real-time ML system — Thompson Sampling allocation, time-series ROAS forecasting, creative-fatigue prediction, and robust anomaly detection — outperform the manual policy of a five-brand e-commerce operator while remaining compatible with strict per-action operator approval?

FIVE ML SUB-PROBLEMS

- ROAS forecast (regression)
- Placement budget allocation (Beta-Bernoulli bandit)
- Anomaly detection (MAD-modified z-score, 8 metrics)
- Creative-fatigue forecast (survival-style regression)
- Anomaly co-occurrence clustering (unsupervised)

THE DATA LAYER

Meta Marketing API v18+ · Hourly sync · 10 ad accounts · 5 brands · 32 days

Table	Rows	Granularity
ad_campaigns	436	Campaign metadata
ad_insights	1,794	Campaign × adset × ad × date
ad_breakdowns	684	Campaign × date × placement
ad_creative_meta	115	Thumbnail · body · CTA
orders	tens of thousands	Order-derived revenue (30d)
ml_budget_suggestions	100	Bandit audit log

Target y for forecast: $ROAS = \text{order-derived revenue} \div \text{spend (not Meta pixel)}$.

1,794

daily ad-set insights

436

active campaigns

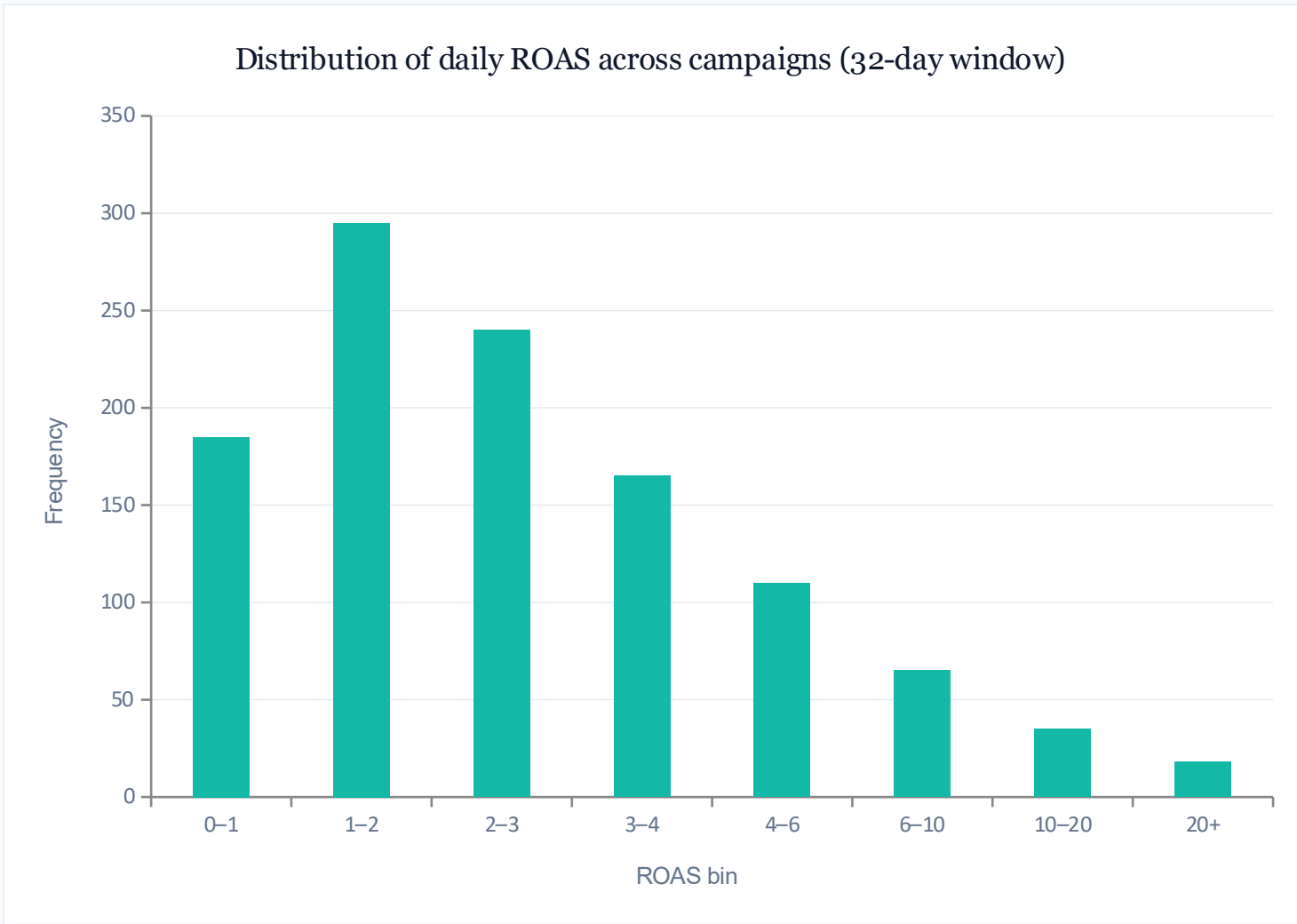
5

brands

24

predictors in X

■ EDA — ROAS Distribution is Log-Normal



KEY STATISTICS

Median	2.63
Mean	2.95
Max	226.78
Skew	right (log-normal)

WHY THIS MATTERS

Classical z-score assumes ROAS is normal. Under right-skewed log-normal data, the empirical σ is inflated by the right tail — masking real downward anomalies, precisely where the operator most needs the alert.

→ Use Iglewicz–Hoaglin modified z-score: σ replaced by $1.4826 \times \text{MAD}$; threshold raised to $|z| \geq 3.5$.

■ Methodology — The ML Pipeline

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STAGE 1

Data Layer

Meta API v18+
Postgres 16
Hourly sync

STAGE 2

Feature Engineering

Lags · MAs · σ
Turkish calendar
Video ratios

STAGE 3

Forecast Ensemble

SARIMA(1,1,1)(1,1,0,7)
XGBoost · OLS
Weights $\propto 1/\text{MAPE}$

STAGE 4

Decision Layer

Beta-Thompson bandit
MAD-z anomaly
Fatigue projection

STAGE 5

Operator Gate

UI approval click
require_admin
Full audit log

WALK-FORWARD VALIDATION

- Train on $[t_0, T-14]$, predict $[T-14, T]$
- Score MAPE · MAE · CI-coverage; roll forward; repeat
- NOT random k-fold — financial data is non-stationary
- Same hold-out used to set ensemble weights ($\propto 1/\text{MAPE}$)

MODEL CARD

Model	Role
SARIMA(1,1,1)(1,1,0,7)	Weekly seasonality + $I(1)$
XGBoost (depth 4, η 0.1)	Non-linear interactions
Linear OLS fallback	Sparse-data backstop
Beta-Thompson bandit	Online budget allocation
MAD-z (Iglewicz-Hoaglin)	Robust anomaly detection

■ Feature Set — 24 Predictors

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Three classes of features capture distinct economic mechanisms: autoregression of past performance, Turkish calendar shocks, and creative-quality leading indicators added in Phase 2.

AUTOREGRESSIVE & MOMENTUM

9 features

- roas_lag1 · roas_lag7
- roas_ma7 · roas_ma14 · roas_std7
- spend_ma7 · spend_change
- clicks_per_spend · cps_ma7

*Yesterday's ROAS conditions today's via auction dynamics.
Spend itself shifts auction position (CPM).*

TURKISH CALENDAR

10 features

- day_of_week · is_weekend
- day_of_month · month · week_of_year
- is_holiday · is_bayram
- is_ramadan_shopping · is_black_friday_week
- is_ecommerce_event · days_to_nearest_holiday

*Salary-cycle effect (Turkish state salaries paid on the 15th).
Bayram and Ramazan shift purchase timing dramatically.*

CREATIVE QUALITY (PHASE 2)

5 features

- hook_rate_ma7 (video_p25 ÷ impressions)
- hold_rate_ma7 (video_p75 ÷ plays)
- video_completion_ma7 (p100 ÷ plays)
- outbound_ctr_lag1
- frequency_lag1

*Leading indicators of creative decay and audience
saturation — direct fatigue signal.*

Validation: time-series walk-forward (train $[t_0, T-14]$ → predict $[T-14, T]$ → roll).

■ Results I — ROAS Forecast · A Deliberate Negative Result

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188.94 %

Walk-forward MAPE

aggregate · pooled

-1570.85

XGBoost R² (validation)

pooled · T = 32 d

T = 32

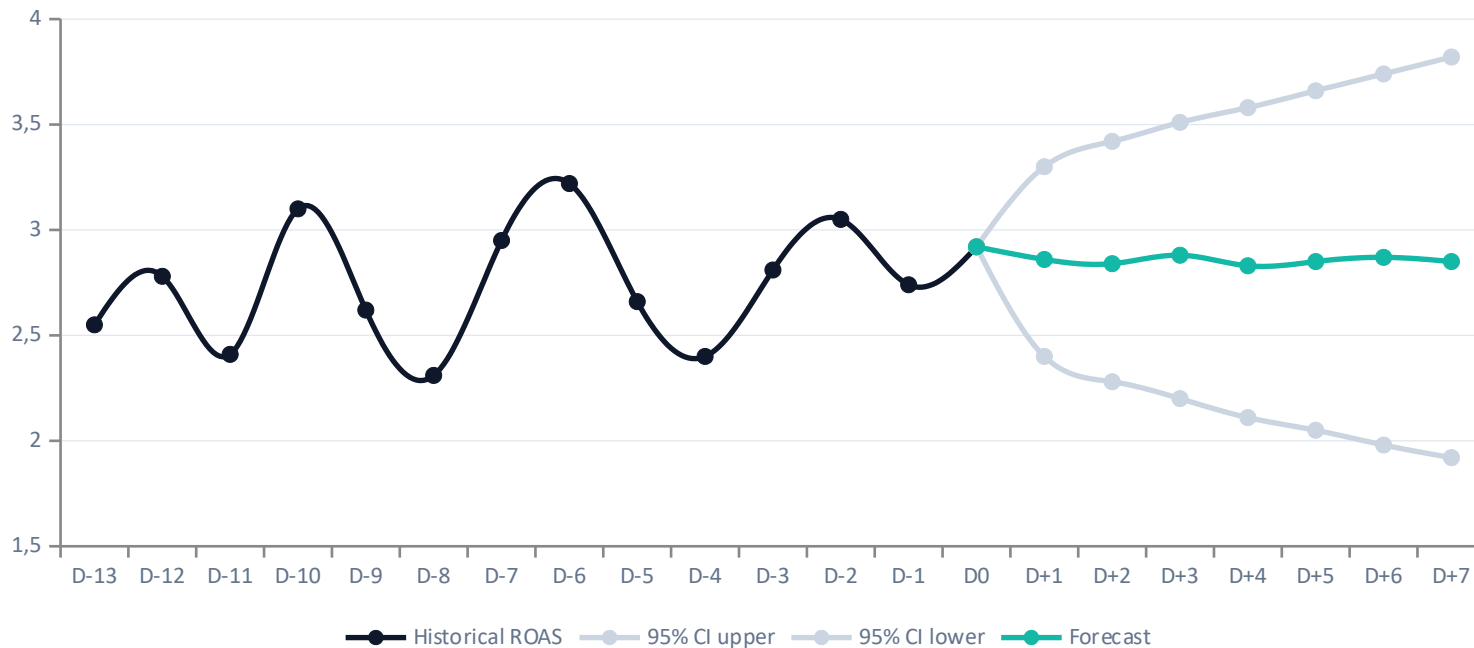
days of history

< 1 salary cycle

per-campaign

models converge
production resolution

SARIMA + XGBoost ensemble — 7-day horizon



WHAT THE FORECAST SAYS

- Aggregate XGBoost — pooling five structurally different brands into one model — fails on walk-forward validation.
- A deliberate negative result: small history (T = 32), concept drift, and pooling loss across brands.
- Production resolution: per-campaign SARIMA + XGBoost, which converge with reasonable CI coverage.
- The negative result is the lesson — more informative for validation design than a marginal positive would have been.

■ Results II — Placement Bandit • The Headline Number

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SINGLE-PLACEMENT WASTE • AUDIENCE NETWORK

≈ 1.8 %

of monthly platform spend — recoverable by pausing one zero-revenue placement (Audience Network).

436

campaigns evaluated

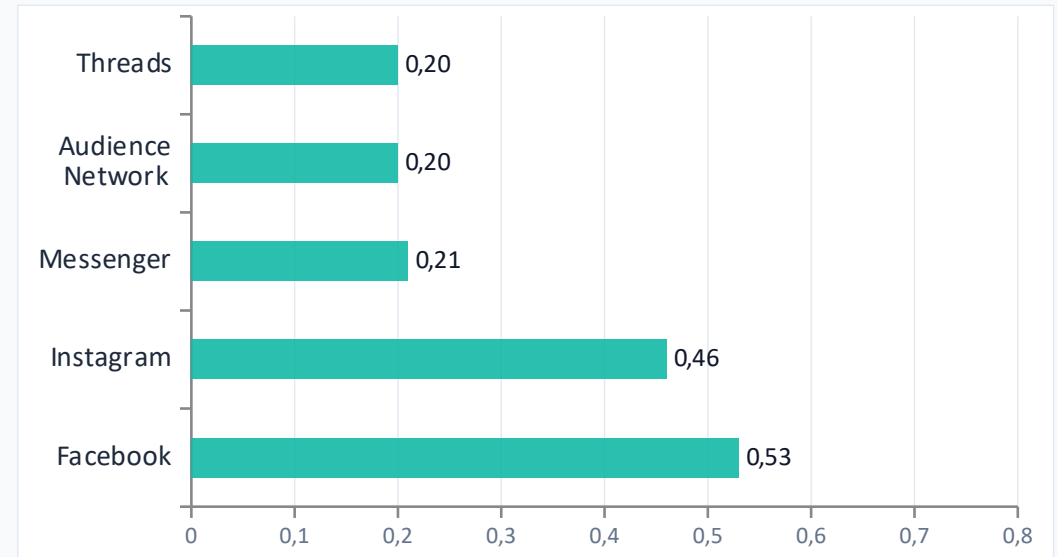
153

suggestions emitted

50

"pause" recommendations

BETA-THOMPSON POSTERIOR • P(success) per placement



CASE • AUDIENCE NETWORK

≈ 1.8 % of spend • 0 revenue

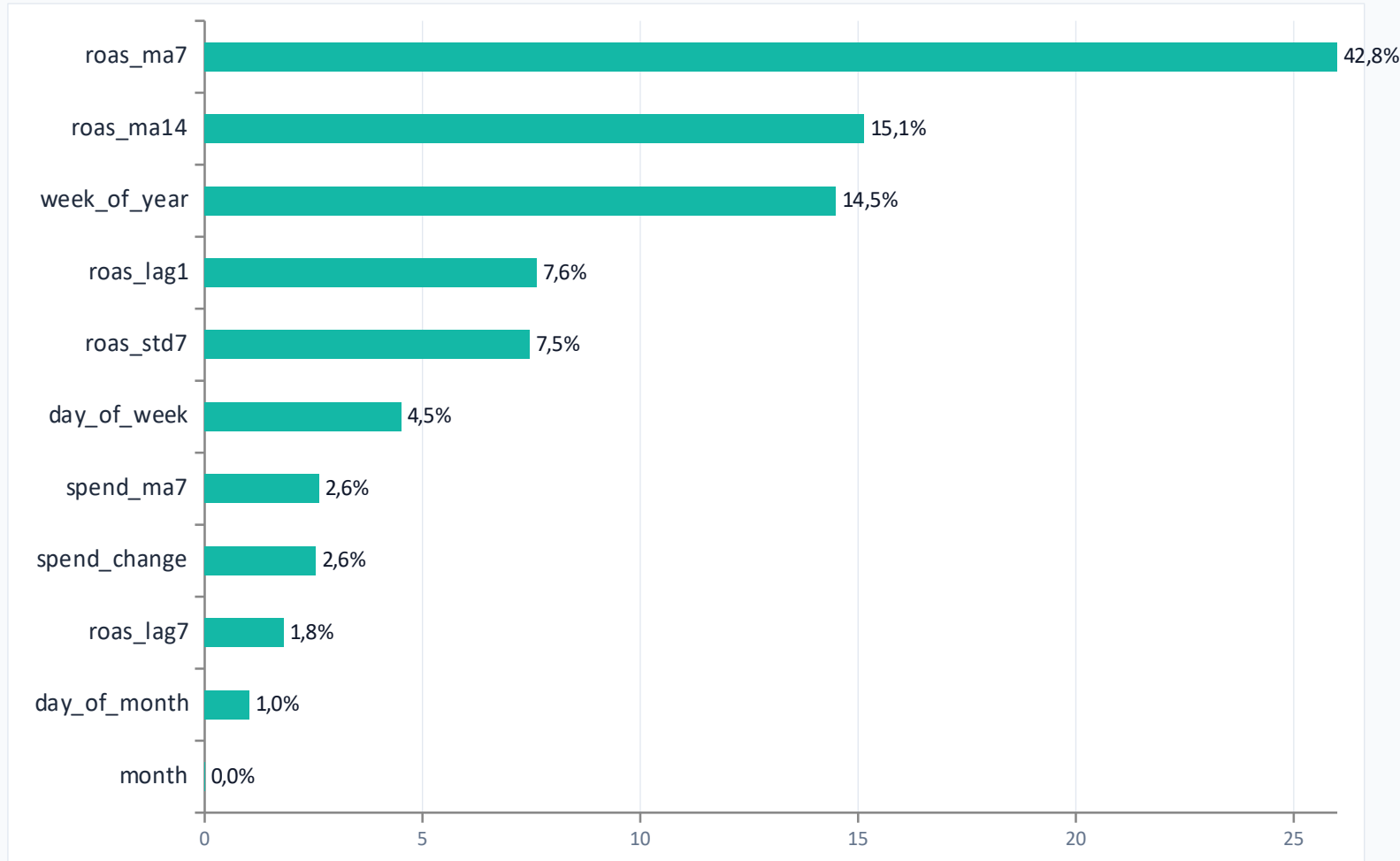
0 attributed revenue across 45 pulls of 15 arms. Beta(15, 60) → P(success) = 0.20 (prior floor).

Concentrated in a single top-spending CPAS campaign arm.

■ Explainability — XGBoost Feature Importance

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Gain-based feature importance from the aggregate XGBoost model, normalized across 24 predictors.



ECONOMIC INTERPRETATION

Autoregression dominates

roas_ma7 alone explains 42.79 % of total gain; the smoothed-lag family (ma7/ma14/lag1/lag7/std7) sums to 73.5 %.

Calendar beats raw recency

week_of_year ranks third (14.49 %), ahead of raw roas_lag1 (7.62 %) — salary cycles, Ramazan and bayram are quantitatively meaningful.

Grouped-attribution caveat

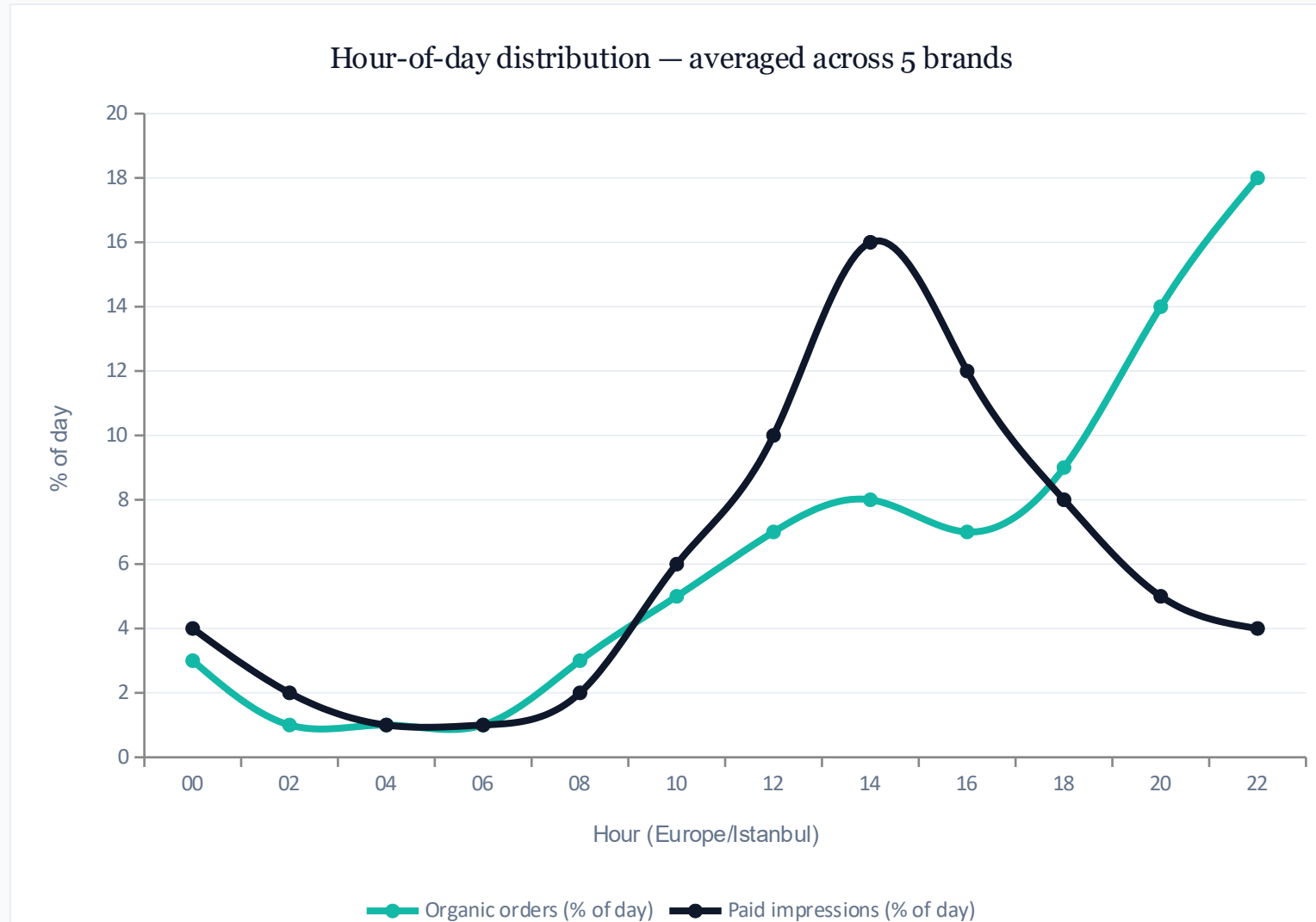
Correlated smoothed lags share signal — read 42.79 % as 'smoothness wins', not 'ma7 alone'.

Phase-2 video features absent

True result, not bug. Creative-quality variation matters per-campaign, not in the aggregate pool. Analogous to inflation_expectations at country vs. global level.

■ Insight — Bimodal Organic Order Pattern

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THE OBSERVATION

Bimodal

organic order pattern — a lunch peak and an evening peak across both product lines.

Lunch peak 10:00–14:00

Evening peak 18:00–22:00

Product line A skews to lunch; Product line B skews to evening.

→ Dayparting bid +30–50 % in BOTH lunch (10–14h) and evening (18–22h) windows. Est. ROAS uplift 5–8 % at constant spend.

■ Policy Implications — Three Actions for the CFO

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01 Audience Network — scheduled withdrawal

FINDING: $\approx 1.8\%$ of platform spend · 0 attributed revenue · $\text{Beta}(15, 60) \rightarrow P(\text{success}) = 0.20$ (prior floor)

ACTION: Pause Audience Network across all 15 flagged arms. Recovery is one click in the same UI if performance proven.

$\approx 1.8\% / \text{mo}$

of monthly platform spend (single placement)

02 Dayparting — shift bid intensity to organic peak

FINDING: Bimodal organic orders — lunch peak 10–14h and evening peak 18–22h

ACTION: Apply hourly bid +30–50 % in BOTH lunch and evening windows (Meta Ads Manager → ad-set schedule).

+5 to 8 %

ROAS uplift at constant spend

03 Creative rotation — pre-emptive replacement

FINDING: 10 ad sets on fatigue watch · highest-confidence forecast: 40.9 days to threshold ($r^2 = 0.97$)

ACTION: Brief creative team to prepare replacement assets for top-10 watch list ahead of fatigue threshold.

Avoid 20–30 %

historical ROAS collapse at frequency > 4.0

All three actions are operator-gated: ML displays the recommendation with full reasoning; the CFO clicks approve before any Meta API mutation.

LIMITATIONS

Sample size & non-stationarity

32 days < 1 full salary cycle + seasonality. CI widens beyond 7 days; we report 14-day but recommend ≤ 7 -day operator weight.

Attribution noise

Meta pixel vs. order-derived revenue diverge 20–40 % daily. Placement-level revenue still depends on pixel — no alternative at sub-campaign granularity.

Right-censoring in fatigue dataset

Only currently running campaigns are observed; operator-paused are right-censored. A Cox proportional-hazards model would replace linear extrapolation.

Black-swan events

Meta algorithm changes, lira devaluation, political events — none in the feature set. The model fails silently on regime change.

CLOSING REMARK

The textbook tools of applied ML — autoregressive time-series, gradient-boosted trees, Thompson sampling, robust outlier detection — translate cleanly into a production ad-allocation system.

The largest single insight was not algorithmic but informational: a single defensible counterfactual — a placement at the prior floor with zero reward — surfaced to a decision-maker who already had the authority to act.

Largest economic value ($\approx 1.8\%$ of monthly spend, a single placement) came from the simplest decision: pause Audience Network.

STACK & REPRODUCIBILITY

FastAPI · PostgreSQL 16 · SQLAlchemy · APScheduler · XGBoost 1.7 · statsmodels SARIMAX · React + TypeScript + Recharts · Alembic 0001 → 0016

Thank you. Questions?